

Task 20 — Dynamic Dashboarding & Storytelling

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Mastery · Expert

Level 20 of 20 · ramp 95%

1 · What this is & why it matters

Build the capstone: a dynamic, interactive dashboard plus a polished data story that ties Phase 1 together.

This is the showcase -- it proves you can go from raw data to a decision-grade, self-serve product with a clear narrative.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A working interactive dashboard plus a narrated data story, demoed live on real data.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Interactivity

filters, drill-downs and parameters that let users explore.

Narrative + dashboard

guided story plus free exploration, together.

Performance under interaction

staying fast as users slice.

Trust signals

definitions, sources and freshness visible.

Decision focus

every view answers a real question.

5 · Recommended stack & tools

- Streamlit / Plotly Dash or a BI tool
- the Phase-1 cleaned data + metric dictionary

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Define the headline story and the key interactions users need.
2. Assemble the dashboard from validated metrics with filters/drill-downs.

3. Layer a guided narrative over the free exploration.
4. Optimise so interaction stays responsive.
5. Add definitions, sources and freshness throughout.
6. Rehearse a live walkthrough end-to-end on real data.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Code dashboard (Streamlit/Dash) vs BI tool.
- Single capstone app vs a deck + linked dashboard.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- What's the one decision this whole thing should drive?
- Where will a live audience try to break it, and are you ready?
- What did Phase 1 teach you that this should showcase?

9 · Definition of Done

Done when all are true and demoable

- A working interactive dashboard plus a narrated data story, demoed live on real data.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A working interactive dashboard plus a narrated data story, demoed live on real data.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Interactivity for its own sake.
- A pretty dashboard with no decision.
- Breaking under the first unexpected filter combination.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design

- Time-series and segmentation techniques